Shor's algorithm (Shor '94)

(BMath Meetup, 09/11/26, ≈45 minutes)

Virtually all of today's internet traffic is secured by schemes based on number theory (elliptic curves & integer factorisation). As an example, we focus on integer factorisation (RSA) cryptography, which is based on the fact that while multiplying numbers is easy to do, factorising them is hard on a classical computer. For example:

$619 \cdot 647 = 400493$ but $400891 = ? \cdot ?$

↑ calculator

$(503)(797)$

There are no known fast algorithms to factorise a given $N \in \mathbb{N}$ on a classical computer. One way to go about it:

1. Pick $1 < a < N$ at random.

2. Compute $\gcd(a, N)$ using Euclid's algorithm (this is fast).
   If $\gcd(a,N) = d \neq 1$, then $d \mid N$. Proceed with $\frac{N}{d}$.
   If $\gcd(a,N) = 1$ (much more likely!) then $a \in (\mathbb{Z}/N\mathbb{Z})^*$, the group of units of integers mod N. In particular, there is a smallest number $\textcircled{r}$ such that $a^{\textcircled{r}} \equiv 1 \bmod N$.

   (Example: $N = 10$, $a = 3$
   $a^1 = 3 \equiv 3$, $a^2 = 9 \equiv 9$, $a^3 = 27 \equiv 7$, $a^{\textcircled{4}} = 81 \equiv 1$)

3. If r is odd, try a different number a.
   If r is even, write $a^r - 1 = (a^{r/2}+1)(a^{r/2}-1) \equiv 0 \bmod N$.
   That is, the prime factors of N appear across $(a^{r/2}+1)(a^{r/2}-1)$.
   Compute $\gcd(a^{r/2}+1, N)$ and $\gcd(a^{r/2}-1, N)$ to find factors of N.
   If $a^{r/2}+1$ happens to be divisible by N, try a different number a.
   (N cannot divide $a^{r/2}-1$ by minimality of N)

The crucial part is **finding r** in Step 2.
Again, there is no known fast way of doing this with a classical computer. Some preliminary comments:

1. The remainders $a^n \bmod N$ repeat periodically with n.
   Example from before: $N = 10$, $a = 3$

$$a^1 = 3 \equiv 3 \;,\quad a^2 = 9 \equiv 9 \;,\quad a^3 = 27 \equiv 7 \;,\quad a^4 = 81 \equiv 1 \;,$$
$$a^5 = 243 \equiv 3 \quad a^6 = 729 \equiv 9 \;,\quad a^7 = 2187 \equiv 7 \;,\quad a^8 = 6561 \equiv 1 \;,$$
$$\dots$$

2. Classical versus quantum computers, bits versus qubits
   A classical bit ☐ exists in one of two states:

   [0] : state $|0\rangle$ (power on)
   [1] : state $|1\rangle$ (power off)

   A sequence of n bits is in one of 2^n states

   [0] [0] : state $|0\rangle$
   [0] [1] : state $|1\rangle$
   [1] [0] : state $|2\rangle$
   [1] [1] : state $|3\rangle$

   This sequence can be measured/read to return its state.
   A quantum bit / qubit, or sequence of such, can be in any superposition of the classical states, for example

$$\tfrac{1}{8}|0\rangle + \tfrac{1}{4}|1\rangle + \tfrac{1}{4}|2\rangle + \tfrac{3}{8}|3\rangle \qquad \left(\tfrac{1}{8} + \tfrac{1}{4} + \tfrac{1}{4} + \tfrac{3}{8} = 1\right)$$

   Measurement returns a classical state according to the distribution.

3. Qubits can be "entangled", meaning the measurement of one influences the state of other (unread) qubits.

The quantum part of Shor's algorithm (sketch).

Pick m s.t. $2^m > N$. Initialise m qubits with the uniform probability distribution of the classical states

$$\frac{1}{2^n}|0\rangle + \frac{1}{2^n}|1\rangle + \frac{1}{2^n}|2\rangle + \dots + \frac{1}{2^n}|2^m-1\rangle$$

Entangle a second set of m qubits with the first one via the associated remainders mod N.

$$\frac{1}{2^n}|a^0 \bmod N\rangle + \frac{1}{2^n}|a^1 \bmod N\rangle + \frac{1}{2^n}|a^2 \bmod N\rangle + \dots + \frac{1}{2^n}|a^{2^m-1} \bmod N\rangle$$

Now, measure this second set of qubits, resulting in a random choice of the above states, say $|\text{rem}\rangle$.

This does not yet tell us anything about the r with $a^r \equiv 1 \bmod N$ but, due to entanglement, the distribution of the first set of qubits changes to be consistent with this measurement, meaning the states $|k\rangle$ where $k \bmod N$ does not equal rem disappear; only the ones where $k \bmod N = \text{rem}$ stay, with equal probability.

(If we had measured the first set, the second would have crystallised to a concrete remainder)

But these states have period r! Example from above: $N=10$, $a=3$

If the remainder measurement was 9, only the states 2, 6, 10, 14, ... remain. Vague but less significant: now apply the quantum Fourier transform to extract the period $r=4$.

For the example $N=400891$ from the beginning with $a=582$. Then $r=1004$.

We compute $\gcd(582^{502}-1, N) = 503$, $\gcd(582^{502}+1, N) = 797$.

Indeed, $503 \cdot 797 = 400891$.